Part II-A parabolic fit can be represented by the equation:

$$y = ax^2 + bx + c$$

The sum of the squared errors is given by:

$$S = \Sigma(y - ax^2 - bx - c)^2$$

This gives us the following normal equations:

$$\Sigma x^4 a + \Sigma x^3 b + \Sigma x^2 c = \Sigma xy$$

$$\Sigma x^3 a + \Sigma x^2 b + \Sigma xc = \Sigma x^2 y$$

$$\Sigma x^2 a + \Sigma xb + \Sigma c = \Sigma xy$$

Part III

$$\Sigma x = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45$$

$$\Sigma x^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 = 285$$

$$\Sigma x^3 = 1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 + 8^3 + 9^3 = 2025$$

$$\Sigma x^4 = 1^4 + 2^4 + 3^4 + 4^4 + 5^4 + 6^4 + 7^4 + 8^4 + 9^4 = 15333$$

$$\Sigma y = 2 + 6 + 7 + 8 + 10 + 11 + 11 + 10 + 9 = 74$$

$$\Sigma xy = (1 * 2) + (2 * 6) + (3 * 7) + (4 * 8) + (5 * 10) + (6 * 11) + (7 * 11) + (8 * 10) + (9 * 9) = 421$$

$$\Sigma x^2 y = (1^2 * 2) + (2^2 * 6) + (3^2 * 7) + (4^2 * 8) + (5^2 * 10) + (6^2 * 11) + (7^2 * 11) + (8^2 * 10) + (9^2 * 9) = 2771$$

Substituting these values into the normal equations, we get:

$$15333a + 2025b + 285c = 1771$$

$$2025a + 285b + 45c = 421$$

$$285a + 45b + 9c = 74$$

We can solve these equations simultaneously to get:

$$a = 0.0017646b = 0.47034c = 0.475$$

Therefore, the equation of the best-fit parabola is:

$$y = 0.0017646x^2 + 0.47034x + 0.475$$